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DATA DESCRIPTOR

BIRDBASE: A Global Dataset of Avian Biogeography, Conservation, Ecology and Life History Traits

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Trait datasets play a crucial role in advancing ecological, biogeographical and evolutionary studies and in prioritizing global conservation efforts. Birds are the best-known major group of organisms and comprise excellent environmental indicators. With their diverse traits, birds offer valuable insights into the provisioning of critical ecological functions and how animals adapt to global change. BIRDBASE is a global avian trait dataset that provides an up-to-date compilation of ecological traits for all birds. With 78 traits from 11,589 bird species in 254 families, and data on 6%–16% more species than the previous avian trait datasets, BIRDBASE includes all the bird species recognized by the four major avian taxonomies. BIRDBASE is also the first avian trait dataset that covers all the species in *AviList*, the new unified global avian checklist. As the most comprehensive avian trait dataset, BIRDBASE will enable new meta-analyses in ornithology, conservation biology and macroecology, expand our ability to understand the origins and maintenance of biodiversity, help quantify avian responses to global change, and inform international conservation efforts.

Background & Summary

Comprehensive and up-to-date datasets of species traits are essential for biogeography, conservation biology, evolution, macroecology and other fields of biological research. Trait datasets enable the integration of a wide range of ecological, morphological, life history and other traits, facilitating comparative studies on organismal ecology, evolution, physiology, behavior, and conservation^{1–10}. With many charismatic flagship species, birds are the best-known major group of organisms, influenced by the historic interest of millions of professional and amateur ornithologists worldwide^{11–14}. Being relatively easy to monitor, and sensitive to environmental factors such as climate, habitat, and food availability¹², birds are also excellent environmental indicators¹⁵. The conservation status of all bird species has been assessed earlier and more frequently than any other taxonomic class¹⁶. These efforts have generated a wealth of long-term data, making birds an ideal candidate group to address critical conservation questions, such as how shared ecological traits result in comparable population trends^{13,14} and in similar predisposition to extinction risk^{17–20}.

Here, we present BIRDBASE, a global avian trait dataset where we compiled data on 78 traits of all bird species (11,589), genera (2,398), families (254), and orders (44), based on the most recent avian taxonomies published in October 2024²¹ (*eBird/Clements v2024*), January 2025¹⁶ (*HBW/BirdLife 9.1*), March 2025²² (*IOC v15.1*), and in June 2025, the first unified global avian checklist (*AviList*)²³. Past avian datasets have focused primarily on aspects of morphology^{1–3,24}, ecology or life history^{4–7,9,10,25}, whereas BIRDBASE covers a wide range of biogeographical, morphological, ecological, behavioral, and life history traits (Fig. 1). Spanning data from 367 sources published between 1957 and 2025, BIRDBASE incorporates essential traits in important categories such as diet, habitat, ecological specialization, elevational limits, breeding behavior, and movement ecology (Fig. 2).

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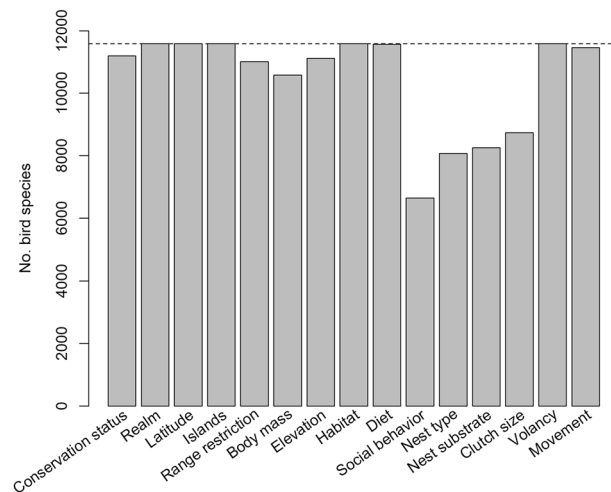


Fig. 1 Trait data for the world's bird species provided in BIRDBASE, a global avian trait dataset. Shown here are 15 categories of bird data contained within BIRDBASE. The height of each bar represents the number of species for which BIRDBASE provides data, the maximum being all the 11,589 avian species (represented with the dashed line).

Consequently, BIRDBASE has been instrumental in addressing a diverse range of key ecological, evolutionary, and conservation questions, including on migratory species^{26–28}, protected area effectiveness²⁹, island biogeography^{30,31}, functional ecology^{32–34}, occupancy dynamics³⁵, drivers of taxonomic discrepancies³⁶, acoustic frequency characteristics³⁷, as well as the impacts of climate change^{38,39}, forest fragmentation^{40,41}, and alien bird species⁴².

With unprecedented taxonomic breadth, BIRDBASE includes between 6% to 16% more bird species than the previous avian trait datasets (depending on the dataset), making BIRDBASE the most comprehensive in the world. BIRDBASE is the only avian trait dataset to include all the bird species recognized by the three principal avian taxonomies (eBird/Clements v2024, HBW/BirdLife v9.1, and IOC v15.1). Since February 2021, the Working Group on Avian Checklists, set up under the auspices of the International Ornithologists' Union, has been working to produce a unified global avian checklist that was published in June 2025 as AviList²³. BIRDBASE is the only avian trait dataset that also covers all the species in this first unified checklist of the birds of the world.

Methods

We compiled the available data on bird species recognized by the three major avian taxonomies^{16,21,22} (eBird/Clements v2024, HBW/BirdLife v9.1, and IOC v15.1), as well as the species recognized in the new AviList, the first unified global avian checklist²³. We collected species-level trait data based on ornithological publications, published datasets, and the first author's field observations of over 9400 bird species. Data were primarily compiled from ornithological sources, including regional books (e.g., *Birds of Africa*⁴³), field guides, specialized datasets^{1,17,44}, and primary literature ([Sources](#) column and [Data Sources](#) worksheet). Most of these data were synthesized from the *Handbook of the Birds of the World*⁴⁵/Handbook of the Birds of the World Alive⁴⁶, now *Birds of the World*⁴⁷, augmented with other sources including BirdLife International¹⁶, field observations, and the ornithological literature. Primary sources from which information was incorporated into BIRDBASE for a given species are noted within the [Sources](#) column in the dataset, with the abbreviations for these sources explained in the [Data Sources](#) worksheet.

Definitions of traits. For each bird, if a species contained multiple subspecies or populations that varied in the range of traits (e.g., different body masses or elevational ranges for various subspecies that are not recognized as species by any of the four main taxonomies), we provide trait information that covers the range of values across the entire breadth of a species' populations (rather than, for example, focusing on the nominate subspecies or the largest population). The [Legend](#) worksheet in the dataset provides additional details on the definitions and scoring approaches for the various traits in BIRDBASE. If every cell for a particular trait or set of traits (e.g., body mass, elevation, social behavior, or movement) for a species is blank, this indicates that we did not find published information on that particular trait for this species.

Conservation status for each species in our dataset is based on the 2024 Red List assessment by BirdLife International¹⁶ and consists of Least Concern (LC), Near Threatened (NT), Vulnerable (VU), Endangered (EN), Critically Endangered (CR), Possibly Extinct (CR (PE)), Possibly Extinct in the Wild (CR (PEW)), Extinct (EX), Extinct in the Wild (EW) or Data Deficient (DD). Since BIRDBASE includes all the recognized species that are known to have gone extinct since 1500 CE, it complements a recently published dataset on extinct and probably extinct species¹⁷, and includes trait data on additional poorly known and rarely encountered CR or DD species, such as New Caledonian Rail (*Gallirallus lafresnayanus*), Kinglet Calyptura (*Calyptura cristata*) and Red Sea Swallow (*Petrochelidon perditia*), that were not treated in that dataset.

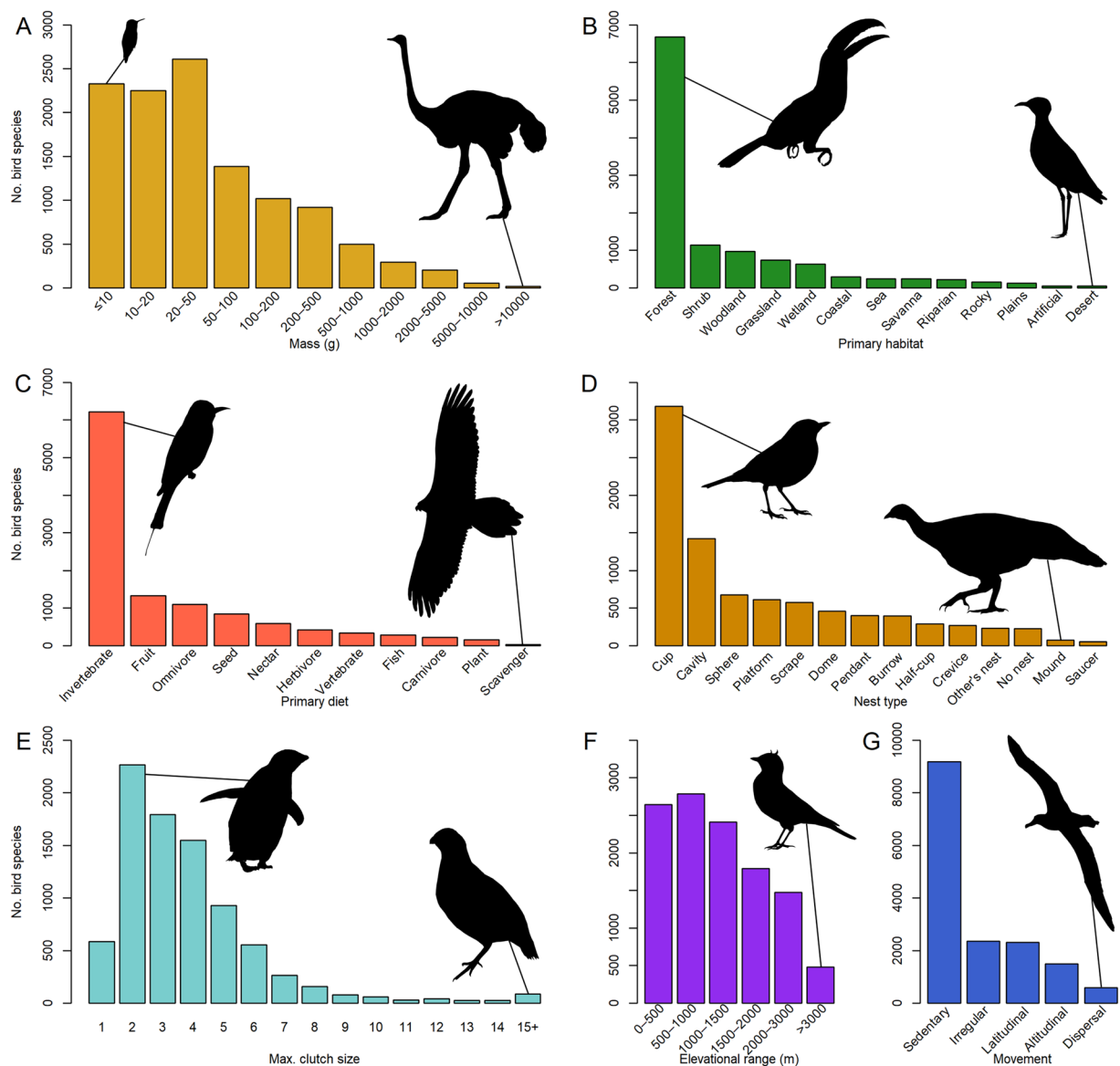


Fig. 2 Variation in seven example traits provided in BIRDBASE, a global avian trait dataset. Bars indicate the number of species within categories or subsets of each trait: (A) average body mass, (B) primary habitat, (C) primary diet, (D) nest type, (E) maximum clutch size, (F) elevational range, and (G) movement strategy. For movement, partial and full latitudinal migrants are grouped together. Each bar for nest type and movement represents the total number of species that either utilize that nest type or can undergo that particular movement strategy (i.e., if a species has more than one type of nest or movement, it is counted separately for each category). Example species (from left to right, top to bottom) include: Bee Hummingbird (*Mellisuga helenae*), Common Ostrich (*Struthio camelus*), Keel-billed Toucan (*Ramphastos sulfuratus*), Cream-coloured Courser (*Cursorius cursor*), Rainbow Bee-eater (*Merops ornatus*), Bearded Vulture (*Gypaetus barbatus*), Chinese Blackbird (*Turdus mandarinus*), Malleefowl (*Leipoa ocellata*), Adelie Penguin (*Pygoscelis adeliae*), Northern Bobwhite (*Colinus virginianus*), Horned Lark (*Eremophila alpestris*), and Snowy Albatross (*Diomedea exulans*).

Location variables include biogeographical realm, latitudinal distribution, range restriction, and whether the species' breeding range is limited to islands. We define an island as a landmass that is surrounded by water and is smaller than a continent, with Greenland treated as the world's biggest island and Australia as the smallest continent. In addition to traditional realms such as Indomalaya, Nearctic, Afrotropics^{48,49}, etc., the biogeographical realm column (Fig. 3) also includes more localized categorizations such as New Zealand and Wallacea⁵⁰. Latitudinal distribution is categorized in three groups: tropical (species' entire range lies between the Tropics of Cancer and Capricorn), temperate (species' entire range lies outside the tropics), and trans-latitude (species range covers both temperate and tropical regions). As with biogeographical realm, these latitudinal classifications are based on species range maps. Following BirdLife International¹⁶, we include a range restricted ("RR") column that scores species with a global range size of less than 50,000 km² with 1 and those with a larger range size with 0.

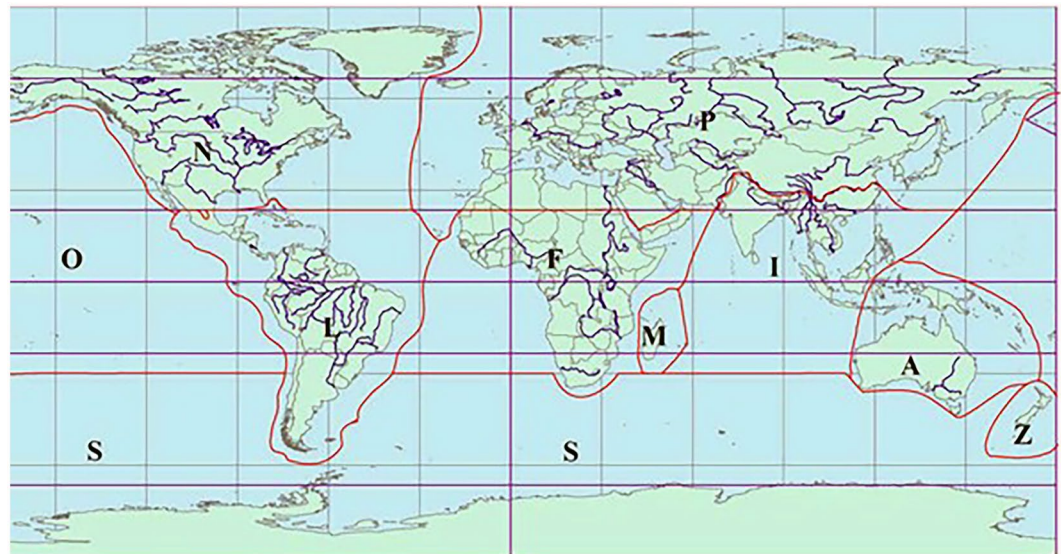


Fig. 3 A map of the world's biogeographical realms used in BIRDBASE, a global avian trait dataset. Red lines circumscribe each realm: A = Australian; F = Afrotropical and sub-Saharan continental Africa; I = Indomalayan; L = Neotropical; M = Madagascar and nearby islands; N = Nearctic; O = Oceania; P = Palearctic; S = South Polar; Z = New Zealand and nearby islands. Note that Wallacea is not depicted on this map but is situated between the Australian and Indomalayan realms. Purple lines indicate the Prime Meridian (vertical) and, latitudinally from top to bottom, the Arctic Circle, the Tropic of Cancer, the Equator, the Tropic of Capricorn, and the Antarctic Circle. Reprinted with modification from Şekercioğlu, Ç.H., Daily, G.C., Ehrlich, P.R. 2004. Ecosystem consequences of bird declines. *PNAS* **101**: 18042-18047. Copyright (2004) National Academy of Sciences.

Body mass of each species is given in grams (g) and separated into columns by sex: Female, Male, and Unsexed (noted when the masses listed in the literature are not distinguished as pertaining to a specific sex). For each sex, “MinMass” is the lower limit while “MaxMass” is the upper limit for that species (if only one value is listed in the literature, this is noted under the lower limit for a species). We excluded values that were described as rare or pertaining to emaciated or malnourished individuals of a species but included values (when available) that were across the different periods of a bird's annual life cycle to best reflect the variation in a species' mass in a given year. We also checked our mass values for all species against those published in the *Handbook of Avian Body Masses*¹ and its update⁴⁴, and incorporated any values into our dataset that were either outside of our sex-specific mass ranges or for which species had no prior masses; a “D07” and/or “D23” is noted in the **Sources** column for these species. For “average body mass” (Fig. 2A), up to six values across the six sex-specific columns were averaged.

Elevational distribution includes the typical or main upper and lower elevational limits of each species' center of abundance (“NormMin” and “NormMax”), as well as either extreme values (rare occurrences) or uncommon elevational limits (“ExMin” and “ExMax”); if elevational distribution limits were not specified as typical or atypical, values were listed under the normal limits. If one of a species' elevational limits is not specifically noted in the literature in meters above sea level but is characterized as being either lowland (L), foothill (F), or montane (M), then we note this instead with the respective abbreviations. We entered a 0 for any species whose range included sea level, mangroves, or marine coastal areas. Our treatment of elevation for species is both more comprehensive and includes more species than do other datasets^{16,25} and also provides corrected limits for some species' published elevational distributions¹⁶. **Elevational range** is calculated as the difference between a species' NormMin and NormMax elevational limits (Fig. 2F).

Main **habitat types** are forest (F), bamboo (BM), woodland (WD), shrubland/scrubland (SH), savanna (SV), grassland (G), plains (PL), rocky (R), desert (D), artificial (A), riverine/riparian (RV), coastal (C), wetland (W), open sea or pelagic (SE), and other (O); for the latter, a description of this habitat (e.g., coral atolls) is noted in the “O-DESC” column. We therefore categorized all habitat descriptions into these types; for example “marsh”, “bogs”, or “fens” would all be classified as wetland (see the data *Legend* for all examples within habitat types). Habitats are numbered in order of preference, as noted in the literature, with **primary habitat** being the main habitat type utilized by a species (Fig. 2B). We calculated **habitat breadth** as the sum of different number of habitats used by each species.

Diet types were classified based on eight major food categories (invertebrates (IN), fruits (FR), nectar (NE), seeds (SE), tetrapod vertebrates (VE), fish (FI), scavenger (such as carrion or garbage; SC), non-reproductive plant material (PL), and miscellaneous (such as vertebrate eggs or beeswax, described in the “Desc” column; MS). We converted qualitative data on relative diet categories into a quantitative form. We assessed scores of use from 0 (non-use) to 10 (exclusive use) for each species, with scores summing to either 0 (for species with no known diet and an interpolated diet could not be quantified) or 10 per species over all categories, thus providing a standardized, probabilistic assessment of species' dietary preferences⁴. If exact diet composition percentages

were available, we used those (e.g., 37% equals a score of 3.7). Most expert descriptions in the literature, however, do not provide numerical details about the proportional use of specific diet categories, but they are usually sufficiently detailed for proportional allocation. To ensure consistency, we applied a standardized approach for interpreting specific terms and the corresponding quantities they represent. In the case that a species was noted as very infrequently or “rarely” feeding on a food source, we denoted that with a “T” for trace feeding. If a food source was described as “occasionally” being consumed, we designated this with a weight of 1; for more information on the protocol for weighted scores, see the *Legend* of the dataset. When numerous diet items are identified as important, we lowered each score accordingly. For instance, if a species is described as mostly eating insects and sometimes consuming fruits and nectar, invertebrates would receive a score of 6, while fruits and nectar would each be assigned 2. If only one item followed “sometimes”, the first received a score of 8 and the second was scored 2. In species diet descriptions, the most important categories were listed first. Therefore, when comparative terms were not used in the diet description, we gave the first-listed category a higher score, with progressively smaller proportions for those that followed. We adjusted the scores based on the overall context and the details of the diet account.

If a diet column is greater than or equal to 6, that diet type is assigned as the species’ primary diet (Fig. 2C). Otherwise, if none of the weighted categories are 6 or greater, the diet was assigned as “Carnivore” (if the species primarily eats a variety of animal matter, including invertebrates) or as “Herbivore” (if it primarily eats a variety of plant matter) (Fig. 2C). However, to clarify which species are still primarily herbivorous or carnivorous despite having a small portion of animal or vegetative food, respectively, we chose a threshold of 20% (a score of 2). For example, if the sum of plant-based categories is greater than or equal to 8, then the species’ primary diet is classified as “Herbivore” even if it has up to 20% of its diet coming from an animal category. A diet consisting of a relatively equal combination of animal and plant matter, where no food group had a score greater than 5 is assigned as “Omnivore”, including any species with greater than 20% of its diet coming from both animal and plant sources. Additionally, while we strove to designate diet scores for each species based on published information (e.g.,^{47,51}), this was not known for every bird species. Therefore, we interpolated diet scores for these species based on their congeners; we note this in a “Diet_Lit” column, with species with a score of 0 having interpolated diets. BIRDBASE includes diet designations for between ~600–1600 more species than the other available datasets^{2–4}. Diet breadth is calculated as the number of major food groups that a species feeds on (e.g., invertebrates, fruits, and seeds = 3).

Ecological specialization index (ESI) is calculated from a species’ dietary breadth and habitat breadth values⁵². ESI is calculated as $\log_{10}(100/[\text{dietary breadth} \times \text{habitat breadth}])$, with a maximum of 2 for the most specialized species that only feed on one major food group and live in one major type of habitat (e.g., a forest frugivore^{17,52}) and a minimum of 0.018 for the most generalist species that feed on all of the eight major food groups and were recorded from a maximum of 12 major habitat types.

Social behavior refers to the extent to which the individuals of a species tend to associate primarily with conspecifics. It is classified across six “Social_#” columns as follows: 1 = colonial, 2 = social (e.g., large numbers of congregating birds in winter flocks or in association with mixed-species groups), 3 = pairs and family groups, 4 = singly and pairs, 5 = solitary, and 6 = lekking. Zero means that a species is not known to engage in that particular social behavior while 1 means it does. Cooperative breeding was classified as: 0 = not known to cooperatively breed (which is different from there being no information available for a species, indicated by a blank cell) or described as being a solitary breeder, 1 = cooperative breeder, 2 = family member helper (i.e., offspring from a previous brood), and 3 = occasional helper. Monogamy was defined based on a species’ known social arrangement(s) and was classified in two columns as either: “Mono” = monogamous, or “Poly” = polygynous/polyandrous/polygamous/etc. Scores of 1 or 0 were used to designate if a species engaged or did not engage in that type of social arrangement, respectively, including occasional engagement. For two of these variables, if a species engaged in behaviors across multiple categories (e.g., a social species that also lekked, or a largely monogamous species that sometimes engaged in polygyny), then each applicable category was noted.

Nest type consists of 14 classifications (Fig. 2D), of which 10 (e.g., cup, scrape, saucer, cavity, burrow) follow definitions previously used in the ornithological handbooks^{53,54}. Among the other four classifications, we used “no nest” and “dome” (i.e., an enclosed or sometimes oven-shaped nest with a roof over the nest cup and a defined entrance) types that are also similar to those in a recent nest trait dataset⁵, while also identifying birds that co-opt “other bird’s nests” and differentiating “half cup” nests (i.e., nests identified as shallow cups) from regular cup/bowl nests that are bulkier, deeper, and/or have more structure to them. In instances where a species places one nest type within another, we resorted to only using the ‘primary’ nest type that surrounds the nest. For example, if a species makes a cup nest inside a tree cavity or rock crevice, then only “cavity” or “crevice” are noted because these are distinctive types in our dataset that notably differ from a cup nest placed, for instance, in the open on a branch. Nest substrate refers to what location the nest is placed on/in, across 12 classifications of substrate type (e.g., ground, water, tree, shrub, rock, etc.). We recognize three vegetation substrates that have previously been lumped with other categories (bamboo, stump, and cactus), and distinguish between the artificial categories of “building” and “pole”. While some nest trait datasets^{5,9} have been published recently, we included nest data in BIRDBASE (*Nest Details* worksheet) as it has been a component of the dataset since its inception, and we categorize 14 nest types and 12 nest substrates versus nine⁵ or seven⁹ for each trait, thus providing increased specificity for these facets of a species’ nesting biology. Nest parasitism indicates the occurrence of brood parasitism, with the “Para. 1” column noting if a species is a brood parasite while “Para. 2” denotes whether a species is a host of brood parasites. A score of 0 means a species is not known to be a parasite or to be parasitized, while a score of 1 indicates it is known as such. Some species that have a score of 1 in both columns are known to exhibit intraspecific brood parasitism (e.g., egg dumping), since the species is effectively parasitizing itself.

We also provide a series of trait columns about the breeding period itself. “Brd1” and “Brd2” indicate the minimum and maximum number of broods known for a species; we did not include nesting attempts that followed the failure of a previous brood. For clutch size (Fig. 2E), we provide the minimum (“Clutch_Min”) and maximum (“Clutch_Max”) number of eggs laid, ignoring rare clutch sizes or those that might be the result of egg dumping. We also checked the average of our clutch sizes against those for species that were previously covered in an amniote life-history dataset⁶, and included average values from this dataset in BIRDBASE for species for which we lacked clutch information (these values are noted in the Clutch_Max column, and referenced with “MYR” in the Source column); this results in BIRDBASE having clutch information for ~1850 more species than were previously published⁶. Incu_Sex indicates which sex incubates the egg: F = female, M = male, or B = both sexes. For some species where either both sexes incubate or only the female incubates, we noted both designations. For incubation period, “Incu1” and “Incu2” indicate the lower and upper limits of incubation in days. For fledgling period, “Fldg1” and “Fldg2” indicate the lower and upper limits of fledging in days. If the fledgling period was specified in weeks, we converted these periods into days. Breeding success is noted with “BrS1” and “BrS2”, which describe the lower and upper limits of breeding success (the percent of nests fledging young). Finally, productivity is represented with “Prod1” and “Prod2”, which indicate the range of yearly productivity per pair (number of young raised to fledging). Note that since we have specific definitions for breeding success or productivity that may differ from those noted in some species’ accounts, we strove to only include data that aligned with our treatment of these categories.

Volancy is summarized into three categories, centered on the degree of flightlessness in a bird: flightless (“yes”), volant (“no”), or partially flightless (“partial”). For the latter category, species are designated as partially flightless if they are not known to fly but may do so (e.g., Brown Mesite, *Mesitornis unicolor*), likely did not fly (for some extinct species, such as Bermuda Towhee, *Pipilo naufragus*), certain ages do not fly for extended periods of time (see young Giant Coot, *Fulica gigantea*), or if a volant species contains a flightless subspecies (see White-throated Rail, *Dryolimnas cuvieri*). Some birds like ducks (Anatidae) and auks (Alcidae, e.g., Dovekie, *Alle alle*) are temporarily flightless while molting, but we did not incorporate this into our volancy category.

Movement is summarized in five binary variables related to dispersal and migration behavior (Fig. 2G): latitudinal migrant (“Mig”) - yearly, regular distance movements (e.g., Nearctic-Neotropical migrants); altitudinal migrant (“Alt”) - regular movements from high to low altitudes or vice versa based on the seasons or post-breeding dispersal, but also some species that conduct extensive altitudinal foraging; irregular movements (“Irreg”) - nomadic, erratic, irruptive, wandering (but not vagrancy), unstructured or poorly known seasonal, inter-island, and other irregular movements based on either weather patterns, natural disturbances (including floods, droughts, wildfires, etc.), or fluctuations in food availability; long-distance dispersal (“Disp”) - usually once in a lifetime dispersal, after leaving the nest (i.e., post-fledging dispersal); and sedentary (“Sed”) - year-round resident, may conduct some local or minor seasonal movements but otherwise does not migrate. In contrast to the prior work describing bird movements¹⁰, we treat altitudinal movements as a primary migratory behavior rather than grouping it with other movement types. Additionally, while some authorities may treat dispersive movements as migration¹⁰, we consider them as less-structured, irregular movements unless dispersive tendencies are clearly noted to be of juvenile birds (e.g., some seabirds move enormous distances during their first year(s) of life). Latitudinally migratory species are further classified as “Full” or “Partial” migrants, with species of the latter classification either clearly noted in the literature as being partial migrants (e.g., a large portion of an otherwise migratory species is resident) or having only a small portion of a population conducting some degree of latitudinal movements. Species can display multiple types of movement (indicated by a score of 1 or, for partial latitudinal migrants, a score of 2) due to the differences across various populations and subspecies of the same species resulting from factors such as geographic distribution, resource availability, and climate. For example, island populations of otherwise predominantly migratory species tend to be resident and some high elevation populations of species that are either resident or latitudinal migrants in the lowlands may move elevationally for the winter. Our movement categories provide a higher degree of specificity (Fig. 2G) and therefore more comprehensive detail for a species than may be present in other datasets^{2,3,16} (as well as more accurate designations than those that may be found for some species¹⁶) where only a primary movement status is designated, which can mask finer scale variability within the species.

Data Records

BIRDBASE is available at Figshare⁵⁵ <https://doi.org/10.6084/m9.figshare.27051040>.

The dataset is provided in an Excel file, with separate worksheet tabs for trait values, trait definitions, nest details and data sources. The dataset contains one row per species for each of the 11,589 bird species. Additional explanations for the dataset columns are provided in the *Legend* worksheet of the dataset spreadsheet, which also contains both the dataset content and the source list as separate worksheets. We included 78 traits in 10 major categories: conservation, geographic distribution, morphology, elevational distribution, habitat, diet, social behavior, reproductive behavior, demography, and mobility. Traits include conservation status (2024 IUCN Red List¹⁶), latitudinal distribution, biogeographical realm, island endemism, body mass, elevational limits, primary habitat, habitat breadth, primary diet, diet breadth, ecological specialization, sociality, clutch size, nest type and placement, and movement (Fig. 2).

Technical Validation

Data were updated according to the latest avian taxonomies^{16,21–23} published in 2024 and 2025, with cross-referencing of original sources where needed. Each species’ data were entered manually into the dataset by one of the authors or the data entry was overseen by one of the authors. The majority of the data came from the *Handbook of the Birds of the World* (HBW), volumes 1–17⁴⁵, now *Birds of the World*⁴⁷, and also from multi-volume ornithological sources, books on taxonomic families, and primary literature (*Sources* column

and *Data Sources* worksheet). During the creation and analysis of the first version of BIRDBASE³², only the first seven HBW volumes covering non-passerines had been published⁴⁵. The information for all passerine species for the first version³² were obtained from family-level books, regional books (e.g., *Birds of Africa*⁴³; *Birds of South America*⁵⁶; *Birds of Southeast Asia*⁵⁷), or peer-reviewed papers on individual species, listed in the *Data Sources* worksheet. For recent revisions and updates of the dataset, all bird species were updated with information primarily from *Birds of the World*⁴⁷, including species updates through June 2025. Any new information on the newly described species accepted through the October 2024 Clements²¹ update, HBW/BirdLife 9.1 update in January 2025¹⁶, 15.1 IOC update in March 2025²², and AviList²³ in June 2025 were also incorporated. For validation and quality control, and to ensure the accuracy of the dataset, individual species' columns were checked independently by different people, including those who specialized on specific variables, and/or at different times over the years.

Usage Notes

BIRDBASE is maintained and continuously updated with new literature and data by the Şekercioğlu Biodiversity and Conservation Ecology Lab at the University of Utah. The BIRDBASE dataset is user-friendly and is the most extensive and comprehensive avian dataset to date for all bird species, offering a broad range of traits in one place. The data are openly available to researchers without restrictions. We request that users of BIRDBASE cite this publication and inform us of any errors they find in our dataset or any missing data they think should be included in future editions. BIRDBASE is regularly expanded and checked, and any reported errors will be corrected.

Code availability

No custom code is associated with the presentation of or utility of the data described in the manuscript.

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Author contributions

Ç.H.Ş. conceived and developed the original idea, designed the dataset and the workflow, created the first version of the dataset between 1999 and 2003, and supervised this study. K.D.K. managed the most recent versions of the dataset and updated the design, verified and updated the taxonomy, and led the team that updated the current version of the dataset. K.D.K., F.M.M.M., A.D., A.N.B. and N.O. improved the workflow, verified data, and updated the current version of the dataset. E.R.B., J.J.H., J.D.B., J.S. and M.H.C.N.C. oversaw the updating of previous versions of the dataset and added additional data. Ç.H.Ş. and K.D.K. wrote the manuscript, with assistance from M.H.C.N.C. K.D.K. prepared the data files, with assistance from J.J.H. The data plots were made by M.H.C.N.C. All authors collected, curated, and entered data from primary sources, and all approved the final manuscript.

Competing interests

The authors declare no competing interests.

Additional information

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